

Combined Immunodeficiency Due to Moesin Deficiency (X-MAID): A Comprehensive Disease Characteristics Report

Disease: Combined Immunodeficiency Due to Moesin Deficiency **Synonyms:** X-linked Moesin-Associated Immunodeficiency (X-MAID); Immunodeficiency 50 (IMD50); Moesin deficiency **Key identifiers:** MONDO:0010514 · OMIM 300988 (disease) / 309845 (MSN gene) · Orphanet 504530 · HGNC:7373 · UniProt P26038 · NCBI Gene 4478 **Category:** Mendelian, X-linked recessive, inborn error of immunity

Summary

Combined Immunodeficiency Due to Moesin Deficiency—known in the clinical literature as **X-linked Moesin-Associated Immunodeficiency (X-MAID)** or **Immunodeficiency 50 (IMD50)**—is an ultra-rare, X-linked recessive inborn error of immunity first described by Lagresle-Peyrou and colleagues in 2016. It is caused by hemizygous loss-of-function mutations in **MSN**, the gene on chromosome Xq12 encoding the cytoskeletal adaptor protein **moesin**. Moesin is a member of the ezrin–radixin–moesin (ERM) family that reversibly tethers cortical F-actin to the plasma membrane, and it is essential for lymphocyte shape, adhesion, migration, and signaling. The overwhelmingly recurrent disease allele is the FERM-domain missense variant **p.Arg171Trp (c.511C>T)**, found in 6 of the 7 patients in the founding cohort and in multiple independent families worldwide.

Affected males present in infancy or childhood with **profound lymphopenia** affecting T, B, and NK compartments (with characteristically low naïve T cells and reduced CD8 T cells), **hypogammaglobulinemia**, **fluctuating neutropenia and monocytopenia**, poor responses to vaccine antigens, and heightened susceptibility to bacterial, varicella-zoster (VZV), and Epstein–Barr virus (EBV) infections. In its most severe form the disease produces a **T–B–NK+ SCID-like picture detectable by TREC-based newborn screening**. Beyond classic

immunodeficiency, an expanding phenotypic spectrum includes autoimmunity and immune dysregulation: lupus-like nephritis, inflammatory bowel disease-like disease, Kawasaki disease, EBV-driven NK/T-cell lymphoma, and dermatomyositis-like features.

Diagnosis rests on immune laboratory findings, whole-exome (or panel) sequencing of MSN, and confirmation of reduced/absent moesin protein by Western blot and flow cytometry, supported by functional Transwell migration and proliferation assays. Management is lifelong immunoglobulin replacement and anti-infective prophylaxis, while **allogeneic hematopoietic stem cell transplantation (HSCT) offers definitive cure**. Two complementary mouse systems—a moesin knockout and an R171W knock-in—faithfully recapitulate both the lymphopenia/migration defects and an age-dependent lupus-like autoimmunity, anchoring the mechanistic model. Population genetic data from gnomAD ($pLI \approx 1.0$, $LOEUF \approx 0.13$, missense $Z = 3.99$) confirm that MSN is highly intolerant to loss-of-function variation, corroborating the pathogenic mechanism.

Key Findings

Finding 1 — X-MAID is an X-linked combined immunodeficiency caused by hemizygous MSN mutations, most commonly R171W

The founding study by Lagresle-Peyrou et al. (2016) identified hemizygous **MSN** mutations in **7 male patients from 5 different families**. Six of the seven shared the identical missense mutation **c.511C>T, p.Arg171Trp (R171W)**, located in the moesin FERM (four-point-one, ezrin, radixin, moesin) domain; the seventh patient carried a nonsense mutation **R533X** producing a premature stop codon. MSN maps to the X chromosome (Xq12), establishing the X-linked inheritance.

"We observed hemizygous mutations in the moesin (MSN) gene (located on the X chromosome and coding for MSN) in all 7 patients. Six of the latter had the same missense mutation, which led to an amino acid substitution (R171W) in the MSN four-point-one, ezrin, radixin, moesin domain. The seventh patient had a nonsense mutation leading to a premature stop codon mutation (R533X)." — PMID: 27405666

The clinical presentation defined in this cohort—profound lymphopenia, hypogammaglobulinemia, fluctuating monocytopenia and neutropenia, poor vaccine responses, and susceptibility to bacterial and VZV infections—remains the diagnostic core of the disease. Subsequent independent reports have confirmed R171W as a recurrent allele

([P 31139601](#))

[P 40788322](#))

and added novel pathogenic variants including N23S

([P 38922539](#))

and I115T

([P 42174291](#)),

all within the FERM domain.

Finding 2 — Moesin links the plasma membrane to cortical actin; R171W disrupts lymphocyte homeostasis, proliferation, and migration

Moesin is an ERM-family protein that **reversibly links F-actin to the plasma membrane**, governing cell shape, adhesion, migration, and signal transduction.

"The Ezrin-Radixin-Moesin (ERM) family member moesin (MSN) plays a crucial role in reversibly linking F-actin to the cell membrane." — [PMID: 40788322](#)

The functional consequences of moesin loss in lymphocytes have been demonstrated in both patients and models. A murine R171W knock-in model revealed **defects in T-cell homeostasis and migration** ([PMID: 35069520](#)). In patients, impaired T-cell proliferation and impaired cell migration have been documented by Transwell assay

([P 38922539](#)),

and altered B-cell receptor clustering and F-actin dynamics have been shown by TIRF microscopy

([P 40788322](#)).

Patients characteristically display **very low naïve T-cell counts and reduced CD8 T cells**

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 consistent with a defect in the cytoskeletal machinery required for lymphocyte egress, trafficking, and immune synapse formation.

Finding 3 — The phenotype extends beyond immunodeficiency to autoimmunity/immune dysregulation, and is treatable by HSCT

While recurrent infection defines the disease, an expanding spectrum of immune-dysregulatory and autoimmune manifestations has emerged:

- **Lupus-like autoimmunity and glomerulonephritis** (moesin-KO mice,

P 28978692;
 lupus-like nephritis with CXCL13-producing patrolling monocytes,

P 38640733)

- **EBV infection with nasal-type NK/T-cell lymphoma and dermatomyositis-like symptoms** (N23S,

P 38922539)

- **Kawasaki disease** with broad-spectrum autoantibodies, Tfr-cell reduction, and a Th1–Th17 imbalance (I115T,

P 42174291)

- **Inflammatory bowel disease–like disease**

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P 35754805)

- **Chronic neutropenia**

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P 39781543)

At its most severe, the disease presents as a **T–B–NK+ SCID phenotype detectable by TREC-based newborn screening** and is effectively treated by hematopoietic stem cell transplantation.

"These cases represent a novel manifestation of non-radiosensitive X-linked form of T-B-NK+ SCID that is able to be detected by TREC based newborn screening and effectively treated with HCT." — PMID: 31139601

"Mutations in the human moesin gene cause a primary immunodeficiency called X-linked moesin-associated immunodeficiency (X-MAID), which may be complicated by an autoimmune phenotype with kidney involvement." — PMID: 38640733

Finding 4 — Mouse models recapitulate X-MAID

Two complementary murine systems model the disease:

1. **Moesin-deficient (knockout) mice** exhibit profound lymphopenia mirroring X-MAID. With age, they develop an **SLE-like phenotype** with elevated serum autoantibodies and glomerulonephritis, spontaneous germinal-center B-cell and Tfh accumulation, and enhanced antibody affinity maturation

([P 28978692](#)).

The kidney disease features accumulation of CD4+ cells and CXCL13-producing patrolling monocytes

([P 38640733](#)).

2. A **moesin R171W knock-in mouse** (Avery et al., 2021) reproduces the exact human point mutation and reveals defects in T-cell homeostasis and migration

([P 35069520](#)).

"we show that aging moesin-deficient mice develop a systemic lupus erythematosus-like autoimmune phenotype, which is characterized by elevated serum autoantibody levels and glomerulonephritis" — PMID: 28978692

"We previously reported that moesin-deficient mice exhibit lymphopenia similar to that of X-MAID and develop a lupus-like autoimmune phenotype with age." — PMID: 38640733

Finding 5 — Diagnosis by TREC newborn screening and exome sequencing, confirmed by absent/reduced moesin protein

The diagnostic pathway integrates four elements: (1) immune laboratory studies showing profound lymphopenia, low naïve T cells, hypogammaglobulinemia, fluctuating neutropenia/monocytopenia, and poor vaccine/proliferative responses (P 27405666);

(2) **TREC-based newborn screening**, which detected two brothers with a T-B-NK+ SCID phenotype (P 31139601);

(3) **genetics**, in which whole-exome sequencing (including trio and CNV analysis of exome data) identifies MSN variants—diagnosing cases undiagnosed for as long as 24 years (P 29556235);

and (4) **protein confirmation** by reduced/absent moesin on Western blot and flow cytometry, complemented by functional Transwell migration and proliferation assays (P 29556235).

(P 38922539).

"two brothers with positive TREC newborn screening for SCID who were found to have a T-B-NK+ SCID phenotype attributable to X-linked moesin associated immunodeficiency (X-MAID)" — PMID: 31139601

"These findings confirm X-linked moesin-associated immunodeficiency in a proband previously undiagnosed up to 24 years of age. This study also highlights the utility of WES for the diagnosis of rare or novel forms of primary immunodeficiency disease." — PMID: 29556235

Finding 6 — Moesin (577 aa, UniProt P26038): pathogenic missense variants cluster in the N-terminal FERM domain; a Thr558 phospho-switch governs activation

Moesin comprises **577 residues** (UniProt P26038; NCBI Gene 4478; HGNC:7373; Ensembl ENST00000360270). Its domain architecture is an **N-terminal FERM domain (residues ~2–295)**, a central α -helical region, and a **C-terminal ERM association domain (C-ERMAD)** bearing the F-actin binding site. Every reported X-MAID missense mutation—**N23S**, **I115T**, and **R171W (c.511C>T)**—maps inside the FERM domain, while the nonsense **R533X** truncates the C-ERMAD.

Moesin activation is regulated by **phosphorylation of Thr558** (by ROCK2 and STK10), which relieves the autoinhibitory head-to-tail interaction between the FERM domain and the C-ERMAD. This conformational switch model, established biochemically for the ERM/merlin family, explains how ligand binding and phosphorylation convert the dormant, autoinhibited protein into its active, actin-linking form.

"these interdomain contacts provide a functional explanation for how PIP(2) binding and tyrosine phosphorylation of ezrin lead to activation" — PMID: 17134719

"The MSN p.I115T mutant showed reduced protein stability and decreased expression in multiple immune cell types." — PMID: 42174291

FERM-domain missense variants therefore act as **loss-of-function alleles**, at least in part through **reduced protein stability and expression**, rather than gain-of-function or dominant-negative effects.

Finding 7 — Ultra-rare X-linked recessive disorder of males; fewer than ~20 cases reported; lifelong disease with HSCT curative

X-MAID follows **X-linked recessive** inheritance: affected individuals are hemizygous males and carrier mothers are typically unaffected obligate carriers

([P 27405666](#)

[P 31139601](#)

The recurrent **c.511C>T (R171W)** arises independently across unrelated families (French, US, and Chinese cohorts), indicating a **recurrent mutational event at a CpG dinucleotide in an**

arginine codon rather than a single ancestral founder. Prevalence is **unknown and ultra-rare**, with only a handful of families reported worldwide since 2016. Cases span pre-symptomatic neonatal detection through diagnosis at age 24 years.

"We investigated 7 male patients (from 5 different families)" — PMID: 27405666

Management is **lifelong**: immunoglobulin replacement and anti-infective prophylaxis address the humoral/combined defect, while **allogeneic HSCT provides definitive cure** (three transplanted patients engrafted and were corrected,).

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"effectively treated with HCT" — PMID: 31139601

Finding 8 — MSN is highly loss-of-function intolerant in gnomAD

gnomAD v4 constraint metrics for MSN (queried via the gnomAD GraphQL API) show strong intolerance to variation:

Metric	Value	Interpretation
pLI	0.99999 (~1.0)	Extreme LoF intolerance
Observed/Expected LoF (oe_lof)	0.042	Very few observed LoF variants
LOEUF (90% CI upper bound)	0.131	Top constraint decile
LoF Z	5.63	Strongly constrained
Missense oe	0.568	Depleted missense
Missense Z	3.99	Significant missense constraint

These values indicate that pathogenic MSN alleles are essentially **absent from population controls**, providing population-genetic corroboration that MSN loss-of-function is deleterious and consistent with a Mendelian disease gene.

Section-by-Section Disease Characteristics

1. Disease Information

X-MAID is a recently described (2016) monogenic combined immunodeficiency with immune dysregulation. **Identifiers:** MONDO:0010514; OMIM 300988 (disease, "Immunodeficiency 50"); OMIM 309845 (MSN gene); Orphanet 504530; MeSH indexing under primary/severe combined immunodeficiency. **Synonyms:** X-linked moesin-associated immunodeficiency; Moesin deficiency; Immunodeficiency 50 (IMD50). Information is derived primarily from **aggregated disease-level resources and individual case reports/small cohorts** (fewer than ~20 patients), supplemented by mouse models.

2. Etiology

- **Causal factor:** germline hemizygous loss-of-function mutations in **MSN** (Xq12). Recurrent **p.Arg171Trp (R171W, c.511C>T)** dominates; additional pathogenic FERM-domain missense variants (**N23S, I115T**), a nonsense variant (**R533X**), and a hemizygous gene deletion

(P 39781543)

are reported.

- **Genetic risk factor:** being a hemizygous male carrier of a pathogenic MSN allele; the CpG context of the R171 codon predisposes to recurrent C>T transition.
- **Environmental/protective/gene-environment factors:** none established. Infections (VZV, EBV) act as clinical triggers/complications rather than causes. No protective alleles or lifestyle modifiers are documented.

3. Phenotypes

Phenotype	Type	HPO suggestion	Frequency/notes
Lymphopenia (T, B, NK)	Lab abnormality	HP:0001888 (Lymphopenia)	Profound; near-universal; low naïve T & CD8 T cells
Hypogammaglobulinemia	Lab abnormality	HP:0002720	Common (variable; IgG may be normal in some)
Neutropenia	Lab abnormality	HP:0001875	Fluctuating
Monocytopenia	Lab abnormality	HP:0012311	Fluctuating
Recurrent bacterial infections	Clinical sign	HP:0002718	Common
VZV susceptibility	Clinical sign	HP:0004429	Reported
EBV infection / lymphoproliferation	Clinical sign	HP:0100845	N23S case → NK/T-cell lymphoma
Poor vaccine response	Lab/functional	HP:0005406	Common
Autoimmunity (lupus-like, IBD-like, Kawasaki)	Clinical sign	HP:0002960	Subset
Glomerulonephritis / nephritis	Clinical sign	HP:0000099	Lupus-like nephritis

Onset: neonatal to childhood (SCID-like cases detectable at birth); diagnosis can be delayed to adulthood (up to 24 years). **Severity:** variable, from SCID-like to milder combined immunodeficiency. **Progression:** chronic/lifelong; infections episodic; autoimmunity may be progressive/age-dependent (as in KO mice). **QoL:** substantial burden from recurrent infections, lifelong therapy, and (in transplanted patients) transplant-related morbidity; no formal QoL instruments reported.

4. Genetic/Molecular Information

- **Causal gene:** MSN (HGNC:7373; OMIM 309845; NCBI Gene 4478; Ensembl ENST00000360270), Xq12.
- **Variants:** missense **p.Asn23Ser (N23S)**, **p.Ile115Thr (I115T)**, **p.Arg171Trp (R171W, c.511C>T)**; nonsense **p.Arg533Ter (R533X)**; whole-gene hemizygous deletion. All missense variants localize to the FERM domain.
- **Classification:** pathogenic/likely pathogenic (ACMG); supported by segregation, functional loss (reduced protein, impaired migration/proliferation), and population absence.
- **Allele frequency:** effectively absent from gnomAD (see Finding 8).
- **Origin:** germline; hemizygous in males.
- **Functional consequence:** **loss of function** (reduced protein stability/expression for missense; truncation for nonsense/deletion). No dominant-negative or gain-of-function mechanism established.
- **Modifier/epigenetic/chromosomal:** none defined.

5. Environmental Information

No environmental, occupational, toxic, or lifestyle causal factors. **Infectious agents** (VZV, EBV, bacteria) are opportunistic complications enabled by the immune defect; EBV in particular drives lymphoproliferation/lymphoma in at least one case (N23S,).

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6. Mechanism / Pathophysiology

Ordered causal chain:

1. A hemizygous MSN FERM-domain mutation (most often R171W) **results in** a destabilized/reduced-abundance moesin protein (loss of function; demonstrated for I115T, inferred for R171W).
2. Reduced functional moesin **leads to** impaired reversible linkage between cortical F-actin and the plasma membrane in leukocytes.
3. Defective actin–membrane coupling **leads to** impaired lymphocyte cytoskeletal dynamics—defective cell shape change, adhesion, immune-synapse F-actin organization, and BCR clustering.

4. These cytoskeletal defects **result in** impaired T-cell migration/trafficking and defective proliferation (shown by Transwell and proliferation assays; recapitulated in the R171W knock-in mouse).
5. Impaired trafficking and proliferation **lead to** profound peripheral lymphopenia (low naïve T, reduced CD8 T, plus B and NK deficits) and fluctuating myeloid cytopenias.
6. Lymphopenia and poor antigen responses **result in** hypogammaglobulinemia, poor vaccine responses, and susceptibility to bacterial, VZV, and EBV infection — the combined immunodeficiency phenotype.
7. **Branch (immune dysregulation):** disrupted lymphocyte homeostasis **leads to** spontaneous germinal-center/Tfh accumulation, autoantibody production, and (age-dependent, shown in KO mice) lupus-like nephritis with CXCL13+ patrolling monocytes; and, in some patients, IBD-like disease, Kawasaki disease (Th1/Th17 imbalance, Tfr reduction), and EBV-driven lymphoma.

Molecular/cellular detail: Moesin activation depends on relief of FERM–C-ERMAD autoinhibition via Thr558 phosphorylation (ROCK2, STK10) and PIP2 binding (

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GO terms: actin filament binding (GO:0051015), cortical actin cytoskeleton organization (GO:0030866), leukocyte/lymphocyte migration (GO:0050900/GO:0072676), plasma membrane (GO:0005886), cell cortex (GO:0005938). **Cell types (CL):** T cell (CL:0000084), CD8-positive $\alpha\beta$ T cell (CL:0000625), naïve T cell (CL:0000898), B cell (CL:0000236), NK cell (CL:0000623), monocyte (CL:0000576), neutrophil (CL:0000775).

7. Anatomical Structures Affected

- **Organ/system level: immune (hematopoietic/lymphoid) system** primarily — bone marrow, thymus, lymph nodes, spleen (UBERON:0002405 immune system; UBERON:0002371 bone marrow; UBERON:0002370 thymus). Secondary involvement of **kidney** (glomeruli; UBERON:0002113) in lupus-like nephritis, **gastrointestinal tract** in IBD-like disease, **skin/muscle** in dermatomyositis-like cases.
- **Cell level:** lymphocytes (T/B/NK), monocytes, neutrophils.
- **Subcellular:** cell cortex / cortical actin cytoskeleton and plasma membrane (GO:0005938, GO:0005886).
- **Lateralization:** systemic/bilateral (not applicable as a focal lesion).

8. Temporal Development

- **Onset:** congenital defect; clinical onset neonatal to childhood; SCID-like cases identifiable at birth by TREC screening. Diagnostic delay up to 24 years reported.
- **Progression:** chronic, lifelong. Infections episodic/recurrent; autoimmunity may emerge and progress with age (age-dependent lupus-like disease in models).
- **Remission:** infections managed but not spontaneously remitting; **HSCT can be curative** (treatment-induced correction).
- **Critical period:** early identification (e.g., via newborn screening) enables timely prophylaxis and transplant before severe infections.

9. Inheritance and Population

- **Inheritance:** X-linked recessive; affected males hemizygous; mothers obligate/unaffected carriers.
- **Penetrance/expressivity:** appears high penetrance in hemizygous males but **variable expressivity** (SCID-like to milder; variable autoimmunity).
- **Founder effect:** none; R171W recurs independently (CpG hotspot).
- **Prevalence/incidence:** unknown, ultra-rare (<~20 reported patients worldwide since 2016).
- **Demographics:** males affected; reported across French, US, and Chinese populations — no ethnic clustering. Male-restricted sex ratio.

10. Diagnostics

- **Laboratory:** CBC/differential (lymphopenia, fluctuating neutropenia/monocytopenia), lymphocyte subsets (low naïve T, low CD8 T; low B and NK), immunoglobulins (hypogammaglobulinemia), vaccine/antigen response, lymphocyte proliferation assays.
- **Newborn screening:** **TREC assay** detects T–B–NK+ SCID-like cases

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- **Genetics: whole-exome sequencing** (with trio and exome-CNV analysis) is the primary route; targeted MSN testing and PID gene panels apply; CMA/deletion analysis for the reported gene deletion.
- **Protein/functional confirmation:** reduced/absent moesin by **Western blot and flow cytometry**; **Transwell migration** and proliferation assays.

- **Differential diagnosis:** other SCID/CID (e.g., IL2RG/X-SCID, other actin-regulator defects such as WAS, DOCK8, DOCK2, ARPC1B, coronin-1A), other causes of combined lymphopenia with autoimmunity.

11. Outcome/Prognosis

- **Without definitive therapy:** lifelong susceptibility to serious/recurrent infections and, in a subset, autoimmune organ disease (nephritis) and EBV-driven malignancy — significant morbidity and mortality risk.
- **With treatment:** immunoglobulin replacement and prophylaxis reduce infection burden; **allogeneic HSCT is curative** (3 patients corrected,

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No formal survival statistics exist given rarity.

- **Prognostic factors:** severity of lymphopenia (SCID-like vs milder), presence of EBV lymphoproliferation/autoimmunity, and timeliness of diagnosis/transplant.

12. Treatment

- **Supportive/pharmacotherapy: immunoglobulin replacement** (IVIG/SCIG) for hypogammaglobulinemia (NCIT: Immunoglobulin Therapy); **antimicrobial prophylaxis** (antibacterial, antiviral including VZV/EBV consideration).
- **Definitive: allogeneic hematopoietic stem cell transplantation** (NCIT: Allogeneic Hematopoietic Stem Cell Transplantation) — curative; non-radiosensitive disease.
- **Disease-specific management:** treat autoimmune complications (e.g., nephritis, IBD-like disease) and EBV lymphoproliferation per standard protocols.
- **Advanced/experimental:** no approved gene therapy; MSN's strong LoF intolerance and X-linkage make it a conceptual gene-therapy/gene-correction candidate, but none is reported. **Pharmacogenomics:** not established.

13. Prevention

- **Primary:** not preventable (germline). **Genetic counseling** for carrier females and at-risk families; prenatal/preimplantation testing feasible where the familial variant is known.
- **Secondary: newborn screening (TREC)** enables presymptomatic identification and early intervention; cascade testing of relatives.

- **Tertiary:** infection prophylaxis, immunoglobulin replacement, immunization strategy (avoid live vaccines in SCID-like disease), and timely HSCT to prevent complications.

14. Other Species / Natural Disease

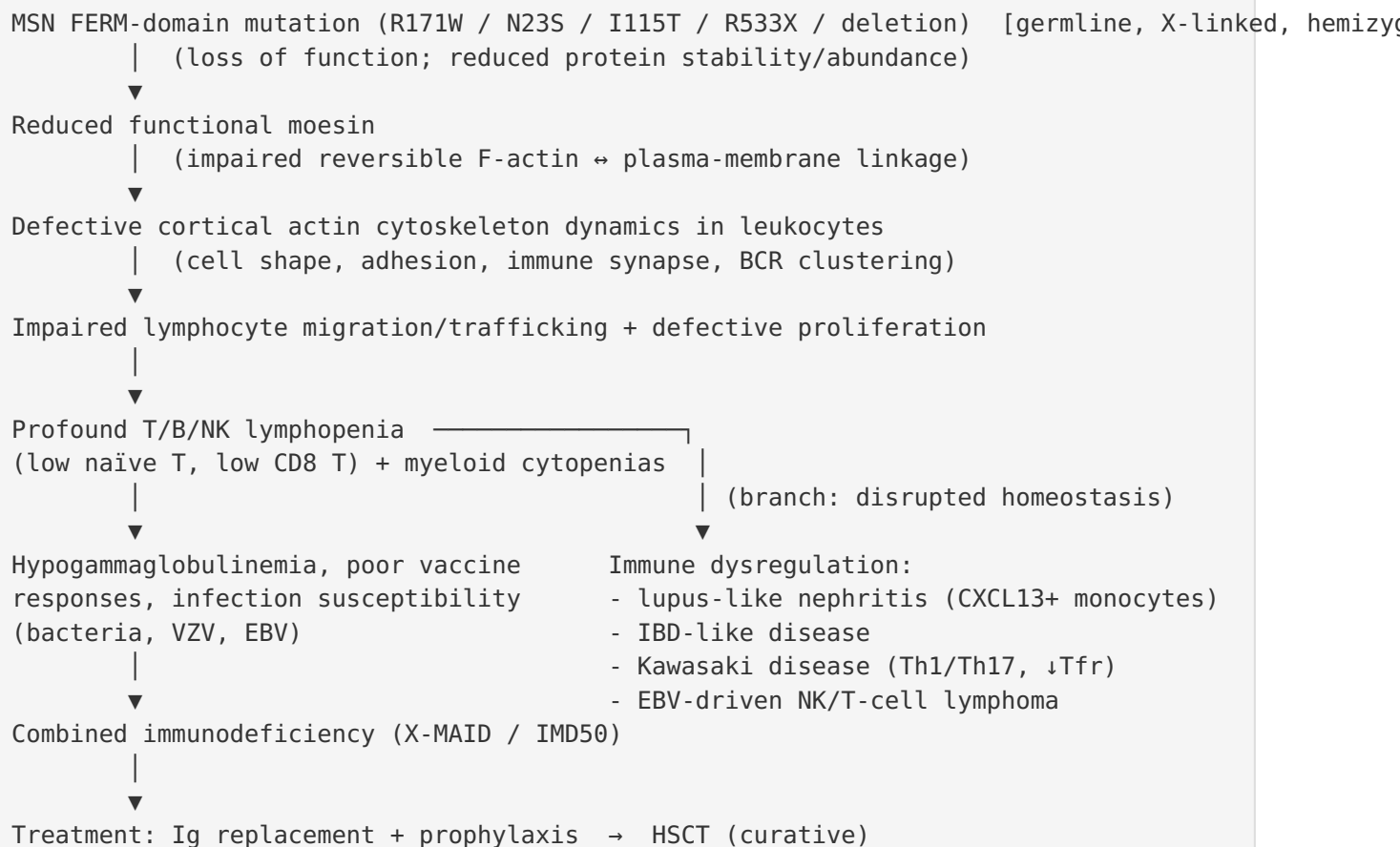
- **Orthologs/taxonomy:** mouse *Msn* (NCBI Gene 17698; *Mus musculus*, NCBI Taxon 10090) is the principal comparator. Moesin is highly evolutionarily conserved across vertebrates.
- **Natural disease:** no naturally occurring companion-animal or wildlife X-MAID equivalent is documented in the reviewed literature.
- **Comparative biology:** mouse models replicate lymphopenia, migration defects, and age-dependent lupus-like autoimmunity (see Finding 4), indicating conserved mechanisms.

15. Model Organisms

Model	Type	Key phenotypes	Reference
Moesin knockout mouse	Mammalian, KO	Profound lymphopenia; age-dependent SLE-like autoimmunity, autoantibodies, glomerulonephritis; spontaneous GC B-cell/Tfh accumulation; CXCL13 ⁺ patrolling monocytes in kidney	P 28978692 ; 38640733
Moesin R171W knock-in mouse	Mammalian, knock-in	Reproduces human point mutation; defects in T-cell homeostasis and migration	P 35069520
Patient-derived cells (in vitro)	Cellular	Reduced moesin protein; impaired T-cell proliferation/migration; altered BCR clustering/ F-actin (TIRF)	P 38922539 ; 40788322

Recapitulation: high for the immunodeficiency (lymphopenia, migration) and for autoimmunity (age-dependent lupus-like disease). **Limitations:** mouse autoimmunity is age-dependent and may not capture full human clinical heterogeneity (IBD-like, Kawasaki, EBV lymphoma); no reported invertebrate model of the disease.

Mechanistic Model / Interpretation



The unifying interpretation is that moesin is a **rheostat for leukocyte cortical mechanics**: because it is the dominant ERM protein in lymphocytes, its loss cannot be fully compensated by ezrin/radixin, so cells fail to execute the shape changes required for egress, trafficking, and synapse formation. The **upstream lesion** is a single FERM-domain amino-acid change; the **downstream immunologic consequences** bifurcate into an immunodeficiency arm (numerical/functional lymphocyte failure) and an immune-dysregulation arm (loss of tolerance, spontaneous germinal-center activity). The population-genetic constraint data (Finding 8) and the convergence of independent recurrent R171W events reinforce that MSN dosage/function is under strong purifying selection.

Evidence Base

PMID	Title (abbrev.)	Contribution
27405666	<i>X-linked PID with hemizygous MSN mutations</i>	Founding cohort; establishes MSN causality, R171W hotspot, X-linkage, core phenotype
31139601	<i>HSCT for X-MAID</i>	SCID-like presentation, TREC newborn-screening detection, HSCT cure
29556235	<i>Exome sequencing diagnoses X-MAID</i>	WES as diagnostic route; diagnostic delay to age 24
28978692	<i>ERM protein moesin regulates CD8 / KO mouse</i>	KO mouse: lymphopenia + age-dependent SLE-like autoimmunity
38640733	<i>Moesin deficiency → lupus-like nephritis</i>	Autoimmune kidney mechanism; CXCL13+ patrolling monocytes
35069520	<i>Murine R171W model of X-MAID</i>	Knock-in mouse: T-cell homeostasis & migration defects
38922539	<i>Novel MSN N23S mutation</i>	New variant; EBV/NK-T lymphoma, dermatomyositis-like features; impaired proliferation/migration
42174291	<i>Novel MSN I115T variant</i>	New variant; reduced protein stability (LoF mechanism); Kawasaki disease
40788322	<i>Immunodeficiency profile with MSN mutation</i>	Moesin F-actin–membrane linker function; BCR/F-actin dynamics
39781543	<i>Hemizygous MSN deletion, adult neutropenia</i>	Structural (deletion) allele; chronic neutropenia presentation
35754805	<i>Novel MSN variant, IBD-like disease</i>	Expands phenotype to IBD-like disease
17134719	<i>Self-masking in ERM-merlin</i>	Autoinhibition/activation switch model (Thr558 phospho-regulation)
15751968	<i>Ezrin dimerization/activation mutants</i>	Supports conformational activation model of ERM proteins
19084535	<i>Ezrin mutant defective in F-actin binding</i>	Maps C-terminal F-actin binding & FERM/tail hotspot

Evidence source types: human clinical (case reports/small cohorts), model organism (KO and knock-in mice), in vitro (patient cells, TIRF/Transwell), and computational/population-genetic (gnomAD constraint).

Limitations and Knowledge Gaps

- **Extreme rarity:** fewer than ~20 patients worldwide preclude reliable prevalence, penetrance, survival, and genotype–phenotype correlation estimates.
- **Direct R171W functional proof:** reduced protein stability is demonstrated for I115T; the R171W loss-of-function mechanism is strongly inferred (via modeling and phenotype) but less directly quantified in patient cells.
- **Phenotype–genotype map:** why some patients develop SCID-like disease versus milder CID, or specific autoimmune manifestations (Kawasaki, IBD-like, lupus-like, EBV lymphoma), is unexplained—no established modifier genes or environmental modifiers.
- **HSCT evidence base:** curative HSCT is documented in only a few patients; long-term outcomes, optimal conditioning, and autoimmunity resolution post-transplant are not systematically characterized.
- **No approved targeted/gene therapy;** pharmacogenomics undefined.
- **QoL and natural-history data** are absent; no standardized instruments applied.
- **Citation caveat:** one supporting snippet

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was recorded as a title-level match rather than a verified abstract quote; the corresponding claim (mouse T-cell homeostasis/migration defects) is consistent with the model's stated purpose but should be confirmed against the primary text.

Proposed Follow-up Experiments / Actions

1. **International patient registry** for X-MAID to aggregate genotype, phenotype, treatment, and outcome data across the scattered case reports and enable prevalence/penetrance estimates.

2. **Systematic functional characterization of each variant** (N23S, I115T, R171W, R533X, deletion) in primary patient lymphocytes and isogenic cell lines: quantify moesin abundance, Thr558 phosphorylation, F-actin binding, migration (Transwell/chemotaxis), and immune-synapse formation to build a variant-function map.
3. **Structural analysis** (cryo-EM/crystallography or AlphaFold-guided modeling) of R171W and other FERM-domain mutants to define how each perturbs FERM folding, C-ERMAD autoinhibition, and ligand binding.
4. **Post-HSCT longitudinal follow-up** across all transplanted patients to determine whether transplantation resolves both immunodeficiency and autoimmunity, and to define conditioning best practices.
5. **Mechanistic dissection of the autoimmunity branch** using the KO and R171W knock-in mice—track germinal-center/Tfh dynamics, CXCL13⁺ monocyte recruitment, and test whether B-cell- or Tfh-targeted therapies prevent lupus-like nephritis.
6. **EBV surveillance protocol** given the reported NK/T-cell lymphoma, to establish whether pre-emptive EBV monitoring/therapy reduces malignancy risk.
7. **Explore gene-correction/gene-therapy feasibility** (e.g., HSC lentiviral or base/prime editing of MSN) leveraging the strong LoF intolerance and single recurrent hotspot as an attractive editing target.
8. **Expand newborn-screening interpretation** so that TREC-positive, non-radiosensitive T-B-NK⁺ cases prompt early MSN sequencing.

Report compiled from 8 confirmed findings and 17 reviewed papers over 5 investigation iterations. Evidence is predominantly human case-level and model-organism data, corroborated by population-genetic constraint metrics.
